Closed-Form Symbolic Raman Pump-Power Controller

Controller definition

The controller maps the 100 input-channel powers S_i (in watts) to six pump launch powers P_k (in watts). The first stage is an affine projection, the second stage gives the six symbolic raw outputs, and the final stage maps them to pump powers. The equations below reproduce the exported symbolic controller.

$$u_j = \sum_{i=1}^{100} A_{ij} S_i + b_j, \qquad j = 1, \ldots, 5. \tag{1}$$

$$P_k = 9.99999974737875 \times 10^{-6} \exp(\mathrm{raw}_k), \qquad k = 1, \ldots, 6. \tag{2}$$

Explicit symbolic output equations

$$\begin{aligned}
\mathrm{raw}_1 = {} & -0.00122745 * u_1^2 + 0.24919218 * u_1 + 0.00324252 * u_2^5 \\
& + 0.00651798 * u_2^4 - 0.03223 * u_2^3 - 0.04849554 * u_2^2 \\
& + 0.14498088 * u_2 + 0.00525884 * u_3^6 - 0.0008007 * u_3^5 \\
& - 0.02866192 * u_3^4 - 0.01129944 * u_3^3 + 0.01789074 * u_3^2 \\
& + 0.07993096 * u_3 + 0.00080161 * u_4^12 - 0.00979341 * u_4^11 \\
& - 0.01479967 * u_4^10 + 0.05821603 * u_4^9 + 0.07290975 * u_4^8 \\
& - 0.11778794 * u_4^7 - 0.14957186 * u_4^6 + 0.07829878 * u_4^5 \\
& + 0.12211469 * u_4^4 + 0.00897673 * u_4^3 - 0.0060414 * u_4^2 \\
& + 0.02012267 * u_4 - 0.00055183 * u_5^2 + 0.26417597 * u_5 \\
& + 6.37817945
\end{aligned} \tag{3}$$

$$\begin{aligned}
\mathrm{raw}_2 = {} & -0.00071734 * u_1^2 + 0.14563063 * u_1 - 0.00337946 * u_2^3 \\
& - 0.00016147 * u_2^2 + 0.0476695 * u_2 - 0.00202889 * u_3^4 \\
& + 0.0060707 * u_3^3 + 0.04689302 * u_3^2 + 0.07261262 * u_3 \\
& + 0.00073952 * u_4^12 - 0.0265746 * u_4^11 - 0.02822968 * u_4^10 \\
& + 0.16331247 * u_4^9 + 0.16730157 * u_4^8 - 0.3418078 * u_4^7 \\
& - 0.38183585 * u_4^6 + 0.23417743 * u_4^5 + 0.32883943 * u_4^4 \\
& + 0.01965891 * u_4^3 - 0.02394043 * u_4^2 + 0.04052616 * u_4 \\
& + 0.16227731 * u_5 + 3.75934142
\end{aligned} \tag{4}$$

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\begin{aligned}
\mathrm{raw}_3 = &-0.00130483 * u_1^2 + 0.26490157 * u_1 - 0.00223721 * u_2^3 \\
&- 0.00373248 * u_2^2 + 0.01708404 * u_2 - 0.00139903 * u_3^6 \\
&- 0.0018639 * u_3^5 + 0.01127127 * u_3^4 + 0.03340034 * u_3^3 \\
&- 0.03110917 * u_3^2 - 0.11149985 * u_3 + 0.00461799 * u_4^9 \\
&+ 0.00387783 * u_4^8 - 0.02282703 * u_4^7 - 0.02304492 * u_4^6 \\
&+ 0.02705668 * u_4^5 + 0.03159498 * u_4^4 - 0.00081815 * u_4^3 \\
&+ 0.00457376 * u_4^2 + 0.02023732 * u_4 + 0.07651051 * u_5 \\
&+ 3.19168083
\end{aligned} \tag{5}
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\begin{aligned}
\mathrm{raw}_4 = &\ 0.09575211 * u_1 - 0.00164903 * u_2^4 + 0.00300732 * u_2^3 \\
&+ 0.01274938 * u_2^2 - 0.00555098 * u_2 - 0.00356661 * u_3^7 \\
&+ 0.00249245 * u_3^6 + 0.00638842 * u_3^5 - 0.0077577 * u_3^4 \\
&+ 0.07139962 * u_3^3 + 0.02275481 * u_3^2 - 0.04886433 * u_3 \\
&+ 0.00691048 * u_4^10 + 0.00755085 * u_4^9 - 0.03372353 * u_4^8 \\
&- 0.03992009 * u_4^7 + 0.04599594 * u_4^6 + 0.06170557 * u_4^5 \\
&- 0.01028314 * u_4^4 - 0.02109516 * u_4^3 + 0.00278603 * u_4^2 \\
&+ 0.00544757 * u_4 + 0.05926517 * u_5 + 1.86688075
\end{aligned} \tag{6}
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\begin{aligned}
\mathrm{raw}_5 = &-0.09958445 * u_1 + 0.00038043 * u_3^5 + 0.00216456 * u_3^4 \\
&+ 0.00251355 * u_3^3 + 0.0162253 * u_3^2 - 0.0494496 * u_3 \\
&+ 0.00368325 * u_4^12 - 0.0232046 * u_4^11 - 0.04994523 * u_4^10 \\
&+ 0.13296939 * u_4^9 + 0.21657785 * u_4^8 - 0.25030801 * u_4^7 \\
&- 0.39883882 * u_4^6 + 0.1441112 * u_4^5 + 0.29564059 * u_4^4 \\
&+ 0.02582485 * u_4^3 - 0.04355886 * u_4^2 - 0.00335631 * u_4 \\
&+ 0.06321963 * u_5 - 0.03287607 * \sin(0.83548572 * u_2 - 9.29860828) \\
&+ 0.1837773
\end{aligned} \tag{7}
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\begin{aligned}
\mathrm{raw}_6 = &\ 0.0011059 * u_1^2 - 0.22451551 * u_1 + 0.00252943 * u_3^6 \\
&- 0.00094261 * u_3^5 - 0.01479762 * u_3^4 + 0.0012466 * u_3^3 \\
&+ 0.05356122 * u_3^2 + 0.23066243 * u_3 - 0.00811534 * u_4^11 \\
&- 0.00884262 * u_4^10 + 0.04695487 * u_4^9 + 0.0542983 * u_4^8 \\
&- 0.09006899 * u_4^7 - 0.11659744 * u_4^6 + 0.05264498 * u_4^5 \\
&+ 0.09169355 * u_4^4 + 0.00903328 * u_4^3 - 0.00662202 * u_4^2 \\
&+ 0.01059174 * u_4 + 0.07680016 * u_5 + 0.0072789 * \sin(0.94357151 * u_2 \\
&+ 0.69923525) - 0.06400169
\end{aligned} \tag{8}
$$

Affine projection coefficients

The matrix entries below are the exported coefficients A_{ij} in $u_j = \sum_i A_{ij} S_i + b_j$. Coefficients are in W^{-1}; the biases b_j are dimensionless.

i	A_{i1}	A_{i2}	A_{i3}	A_{i4}	A_{i5}
1	-12.794230	-51.691760	-33.877460	-37.111510	17.258640
2	-65.334760	-77.732820	-79.500420	-57.875050	101.251800
3	-89.008490	-31.717880	-129.425600	-11.065660	109.884700
4	56.705080	-32.990010	52.289580	-35.273230	-74.991750
5	-53.047730	-57.122220	-59.483290	-45.204750	87.217850

i	A_{i1}	A_{i2}	A_{i3}	A_{i4}	A_{i5}
6	-46.187770	-47.960550	-85.774810	-27.203150	64.794750
7	-9.804624	12.290960	-45.215810	11.858530	-6.513994
8	2.243161	-12.369890	-43.342430	0.907915	-21.380470
9	-26.468430	-20.197510	-29.582550	-14.630040	24.981210
10	15.833200	-22.154170	-25.910800	-19.165610	-26.556120
11	-25.351230	-6.425647	-32.617590	-1.455586	12.112110
12	-5.702503	-42.433860	5.094034	-40.031880	3.255195
13	-4.096383	-34.221170	-35.831340	-22.770030	4.028548
14	49.516790	-55.556840	13.358790	-60.480610	-54.471490
15	-57.431130	-19.201860	-104.285000	-13.989450	76.284160
16	-17.198540	8.175671	-44.585810	20.380280	-8.906894
17	-35.314790	-21.836370	-79.512560	-14.426420	38.134080
18	35.869220	-35.538500	29.565220	-35.198880	-51.138050
19	-15.542570	-46.286630	13.675940	-35.075110	21.513670
20	-54.099170	-63.503160	-70.233970	-55.994580	90.894260
21	-87.068330	-42.079010	-89.137500	-28.553130	120.423600
22	-25.450610	-46.942400	-53.282730	-30.267020	31.848860
23	-2.227203	-16.893220	-19.088530	-14.591120	-11.022330
24	-32.776370	-23.293710	-55.822160	-18.401790	19.158250
25	-19.323250	-15.759540	-53.095610	-15.823980	0.312155
26	76.076930	37.482730	60.237540	20.573770	-143.853900
27	-6.565069	20.951420	-62.982380	9.055981	-22.727020
28	-47.586720	2.077594	-96.021990	-8.746526	34.516790
29	-34.282610	17.010890	-78.050620	21.777770	7.482961
30	-7.086439	-39.249280	-31.636250	-37.297270	18.707900
31	16.888990	-23.089840	15.183760	-31.468910	-35.042240
32	-54.954240	-8.920254	-84.260520	-7.610268	48.512730
33	52.188750	7.661104	27.735300	2.394633	-95.934770
34	-59.540550	-34.642760	-54.557420	-21.400080	74.524020
35	-24.734570	-12.532820	22.238970	-11.513360	14.075430
36	10.557750	-20.380940	4.597333	-15.659390	-24.002730
37	42.217630	-34.184460	63.228890	-37.494300	-57.247910
38	31.895480	-15.418420	41.131790	-15.033080	-53.171240
39	-46.041620	-29.775410	-42.976280	-23.738700	60.271290
40	12.995620	-5.974782	-17.122340	-17.713700	-29.549050
41	73.797100	11.709740	104.289800	-0.566040	-131.229500
42	65.955070	-14.969090	56.404380	-19.998350	-95.381340
43	-40.331930	-98.159470	-14.350240	-91.640220	73.725360
44	38.104090	9.241859	29.435960	9.242436	-77.800250
45	5.380908	-41.719600	2.115041	-41.361610	-1.906625
46	-52.082110	46.303220	-63.055730	55.068950	8.179424
47	28.451610	10.469210	48.361710	11.105250	-69.321560
48	1.572162	11.438440	21.968360	-0.143056	-45.704300
49	-32.111340	-23.530570	-53.063650	-30.992840	37.471500
50	32.995790	2.381106	54.772630	-20.943250	-78.397050
51	-12.410320	29.596450	48.722530	9.518916	-33.609640
52	-98.929940	-37.807800	-64.282480	-49.360400	113.321700
53	-45.538810	-56.636580	26.982340	-67.434960	49.088430
54	-25.715160	15.389000	63.829010	-0.859257	-16.082640
55	12.580320	-0.836057	34.874640	-15.947230	-44.870770
56	-37.456630	-36.042580	-36.307580	-34.566300	49.921000
57	-25.748760	-37.476980	-4.633151	-37.211040	34.272820
58	-42.743020	-58.103910	-0.266661	-56.095630	51.857820
59	-13.882710	-34.308710	23.299470	-35.862890	0.554191
60	39.485240	23.929170	64.707190	15.257690	-84.002310
61	13.040080	3.113444	13.785830	13.336400	-45.122000
62	-89.441440	9.580563	-91.335830	19.141820	93.549970
63	-45.857030	5.068612	-96.362110	4.877663	40.941770
64	-47.513030	-24.521180	-29.498940	-8.052910	56.769100
65	30.100760	39.197200	79.045670	43.930490	-93.492550
66	-20.017210	-36.345580	11.462920	-33.078880	30.557090

i	A_{i1}	A_{i2}	A_{i3}	A_{i4}	A_{i5}
67	-19.557240	-58.685250	-34.160950	-47.389730	40.778240
68	54.718980	0.266476	96.835680	-3.555490	-92.572870
69	-15.548080	-4.589191	-9.753272	-0.534377	-0.171347
70	35.217930	13.576620	23.128490	5.839417	-68.940410
71	-12.045700	-30.713740	25.613620	-36.368730	11.235680
72	-60.405580	-11.374760	-24.695810	-26.741100	56.537190
73	24.928950	-1.897788	84.639390	-16.948610	-65.965870
74	-25.810110	10.812250	14.379280	-8.265585	-2.789612
75	-23.213620	15.670510	55.279370	-7.204509	-11.374640
76	-86.228840	-3.015068	-73.601950	-24.639570	72.368170
77	-37.242900	-21.493150	-19.714430	-52.340450	43.747400
78	-82.291870	-12.599350	-86.336920	-32.691480	93.038270
79	-38.380210	12.265570	-44.099070	0.286880	20.045840
80	-12.784690	43.659010	38.530190	42.153840	-37.464080
81	16.339150	-11.240640	8.174700	-16.552380	-27.163440
82	-22.244960	-56.528560	-44.701530	-41.443080	47.209530
83	-64.518840	-10.680710	-50.639040	-5.568784	74.106310
84	-67.826520	-73.694800	-29.256160	-61.627880	115.311500
85	-60.140130	-55.250800	-87.814600	-39.916310	99.904730
86	-30.480900	-25.111840	-39.510000	-19.313630	52.254540
87	-65.581640	-50.737940	-50.124730	-42.489330	103.909800
88	14.235050	-1.836707	2.304981	-14.196000	-19.933830
89	31.275630	-9.432931	51.375410	-6.391632	-58.728440
90	42.986840	-20.360260	56.566930	-26.692190	-49.949510
91	-65.577130	-45.143090	-64.668030	-47.314850	96.779490
92	-66.403470	-10.121380	-34.070550	-11.723100	58.449700
93	-31.119160	-57.880370	20.638360	-58.849330	49.871230
94	-41.993240	-56.696840	-19.855460	-55.172760	61.606750
95	-62.836590	-64.104790	-81.870410	-59.254630	103.766600
96	-68.976190	-18.168280	-71.013230	-18.541260	91.719770
97	-9.851358	-1.239546	25.791990	8.340735	-19.078340
98	-60.301860	2.546412	-36.799260	-3.443126	61.542100
99	0.074412	34.408640	39.618450	28.982320	-52.386450
100	-10.624100	-35.902140	10.581590	-40.771890	12.112840
b_j	0.913578	0.927746	0.734707	0.977877	-0.548452

Export validation

The symbolic export was compared against the numerical KAN on 1,000 occupancy patterns. The reported relative RMS pump-prediction error was $4.7792747458 \times 10^{-4}$, with a maximum absolute error of $8.0390796949 \times 10^{-6}$ W. These values quantify agreement with the numerical controller on the evaluated patterns.